

Reasoning Model Benchmarking — Full Comparison

Objective

Compare outputs across five prompting strategies while keeping the underlying reasoning problem constant.

Benchmark Question

Five students — A, B, C, D, and E — are standing in a line.

A is standing to the left of B.

C is standing to the right of B.

D is standing to the left of A.

E is standing to the right of C.

Strategy	Output	Result	Main observation
Direct Question	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$	Correct	Minimal context; sufficient for this simple task.
Structured Reasoning	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$	Correct	Explicitly requests reasoning and final answer.
Example-Guided	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$	Correct	Provides a demonstration of the desired reasoning format.
Constraint-Based	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$	Correct	Expresses the relationships symbolically.
Minimal / Output-Focused	$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$	Correct	Keeps only the essential constraints and output request.

Overall Result

All five prompting strategies produced the same correct ordering for this problem: $D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$.

Interpretation

For this single, simple reasoning instance, prompt variation did not change final-answer accuracy. The main difference was the amount and structure of guidance provided to the model.

Limitation

A single question is not enough to establish that one prompting strategy is generally superior. A larger benchmark should contain multiple problem types and difficulty levels and use fixed scoring criteria.

Recommended Next Step

Run the five strategies across 20–50 reasoning questions covering ordering, blood relations, syllogisms, directions, coding-decoding, seating arrangements, and multi-step logic. Record accuracy, reasoning quality, consistency, instruction adherence, and response length.